

2,12

$$\begin{aligned} f &= x_1 x_3 + x_1 \bar{x}_2 + \bar{x}_1 x_2 x_3 + \bar{x}_1 \bar{x}_2 \bar{x}_3 = \\ &= x_1 x_3 (\bar{x}_2 + x_2) + x_1 \bar{x}_2 (\bar{x}_3 + x_3) + \bar{x}_1 x_2 x_3 + \\ &\quad + \bar{x}_1 \bar{x}_2 \bar{x}_3 = \\ &= x_1 \bar{x}_2 x_3 + x_1 x_2 x_3 + x_1 \bar{x}_2 \bar{x}_3 + x_1 x_2 \bar{x}_3 + \bar{x}_1 \bar{x}_2 \bar{x}_3 = \\ &= x_1 x_3 + (x_1 + \bar{x}_1) x_2 x_3 + (x_1 + \bar{x}_1) \bar{x}_2 \bar{x}_3 = \\ &= x_1 x_3 + x_2 x_3 + \bar{x}_2 \bar{x}_3 \end{aligned}$$

(2,2)

$$(x+y)(x+\bar{y})=x$$

$$xx + x\bar{y} + xy + y\bar{y} = x$$

$$x + x(\bar{y} + y) = x$$

$$x + x = x$$

$$x = x$$

(2,4)

(2.4)

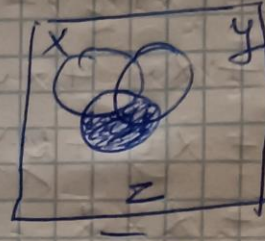
$$xy + yz + \bar{x}z = xy + \bar{x}z$$



xy

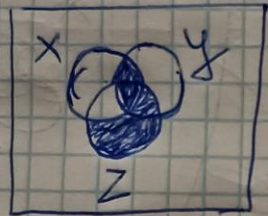


yz



$\bar{x}z$

~~$xy + yz + \bar{x}z$~~



$\equiv$



$$xy + yz + \bar{x}z \equiv xy + \bar{x}z$$